



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 14/07/2026

EXPERIMENT 4 -- DML Commands (UPDATE, DELETE)

AIM:

To perform **DML commands such as UPDATE and DELETE** on a table with constraints.

PROCEDURE:

1. Create a table with required attributes and constraints.
2. Insert records into the table.
3. Modify existing records using the UPDATE command.
4. Remove records using the DELETE command.
5. Display the updated table records using the SELECT command.

PROGRAM:

1. CREATE DATABASE

```
CREATE DATABASE dml_student;
```

```
USE dml_student;
```

OUTPUT:

#	Time	Action	Message	Duration / Fetch
✓ 1	11:30:51	CREATE DATABASE dml_student	1 row(s) affected	0.016 sec
✓ 2	11:31:28	USE dml_student	0 row(s) affected	0.000 sec

2. CREATE TABLE

```
CREATE TABLE student_details (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(50) UNIQUE,  
    department VARCHAR(50),  
    semester INT CHECK(semester BETWEEN 1 AND 8),  
    cgpa DECIMAL(3,2) CHECK(cgpa BETWEEN 0 AND 10)  
);
```

INSERT VALUES:

```
INSERT INTO student_details VALUES  
(101,'Deepthi','CSE',5,8.75),  
(102,'Anu','ECE',4,9.10),  
(103,'Rahul','IT',3,8.50),  
(104,'Priya','CSE',6,9.25);
```

```
SELECT * FROM student_details;
```

OUTPUT:

The screenshot shows a database management tool interface. At the top, a 'Result Grid' displays the contents of the 'student_details' table. Below it, the 'Output' pane shows the 'Action Output' log, detailing the execution of SQL commands and their results.

student_id	student_name	department	semester	cgpa
101	Deepthi	CSE	5	8.75
102	Anu	ECE	4	9.10
103	Rahul	IT	3	8.50
104	Priya	CSE	6	9.25
NULL	NULL	NULL	NULL	NULL

#	Time	Action	Message	Duration / Fetch
2	11:31:28	USE dml_student	0 row(s) affected	0.000 sec
3	11:33:46	CREATE TABLE student_details (student_id INT PRIMARY KEY, student_name VARCHAR(50) UNIQU...	0 row(s) affected	0.062 sec
4	11:34:04	INSERT INTO student_details VALUES (101,'Deepthi','CSE',5,8.75), (102,'Anu','ECE',4,9.10), (103,'Rahul','IT',...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
5	11:34:24	SELECT * FROM student_details LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

3. UPDATE COMMAND:

```
UPDATE student_details  
SET cgpa = 9.00  
WHERE student_id = 101;  
SELECT * FROM student_details;
```

OUTPUT:

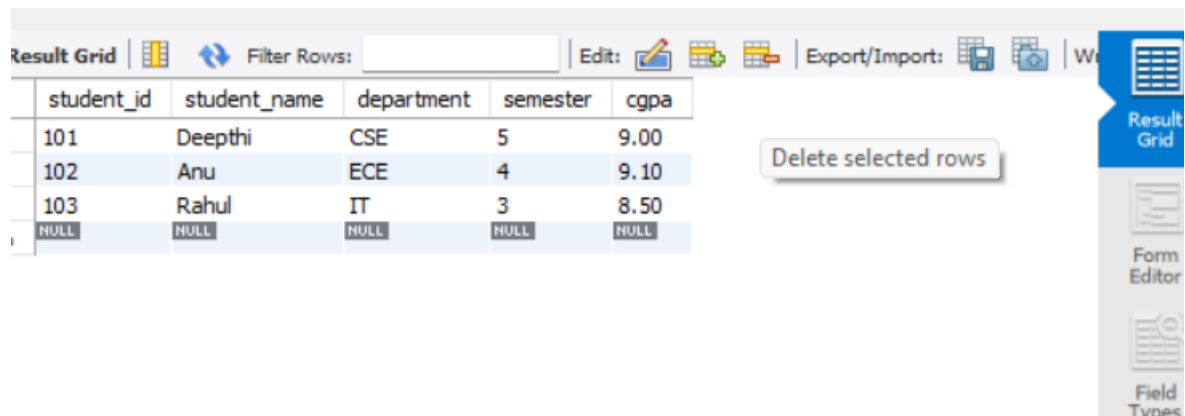
The screenshot shows the 'student_details' table after an update operation. The 'cgpa' value for student_id 101 has been updated from 8.75 to 9.00. The table is displayed in the 'Result Grid' of the database management tool.

student_id	student_name	department	semester	cgpa
101	Deepthi	CSE	5	9.00
102	Anu	ECE	4	9.10
103	Rahul	IT	3	8.50
104	Priya	CSE	6	9.25
NULL	NULL	NULL	NULL	NULL

4. DELETE COMMAND:

```
DELETE FROM student_details  
WHERE student_id = 104;  
SELECT * FROM student_details;
```

OUTPUT:



student_id	student_name	department	semester	cgpa
101	Deepthi	CSE	5	9.00
102	Anu	ECE	4	9.10
103	Rahul	IT	3	8.50
NULL	NULL	NULL	NULL	NULL

RESULT: Data Manipulation Language (DML) Commands such as UPDATE, DELETE are performed